

Task 3: Fault Detection and Recovery Strategy

During nominal operation, the GNSS receiver unexpectedly stops producing measurements while the payload CPU utilisation begins increasing steadily.

Design an FDIR strategy describing:

- Fault detection
- Fault isolation
- Automatic recovery actions
- Transition criteria to SAFE mode
- Conditions for autonomous return to NOMINAL mode

Clearly distinguish between software and hardware faults.

FDIR
strategy
diagram:

Nominal signal loop

Detection
module

Isolation
module

Software fault

Hardware fault

S/W fix module

H/W fix module

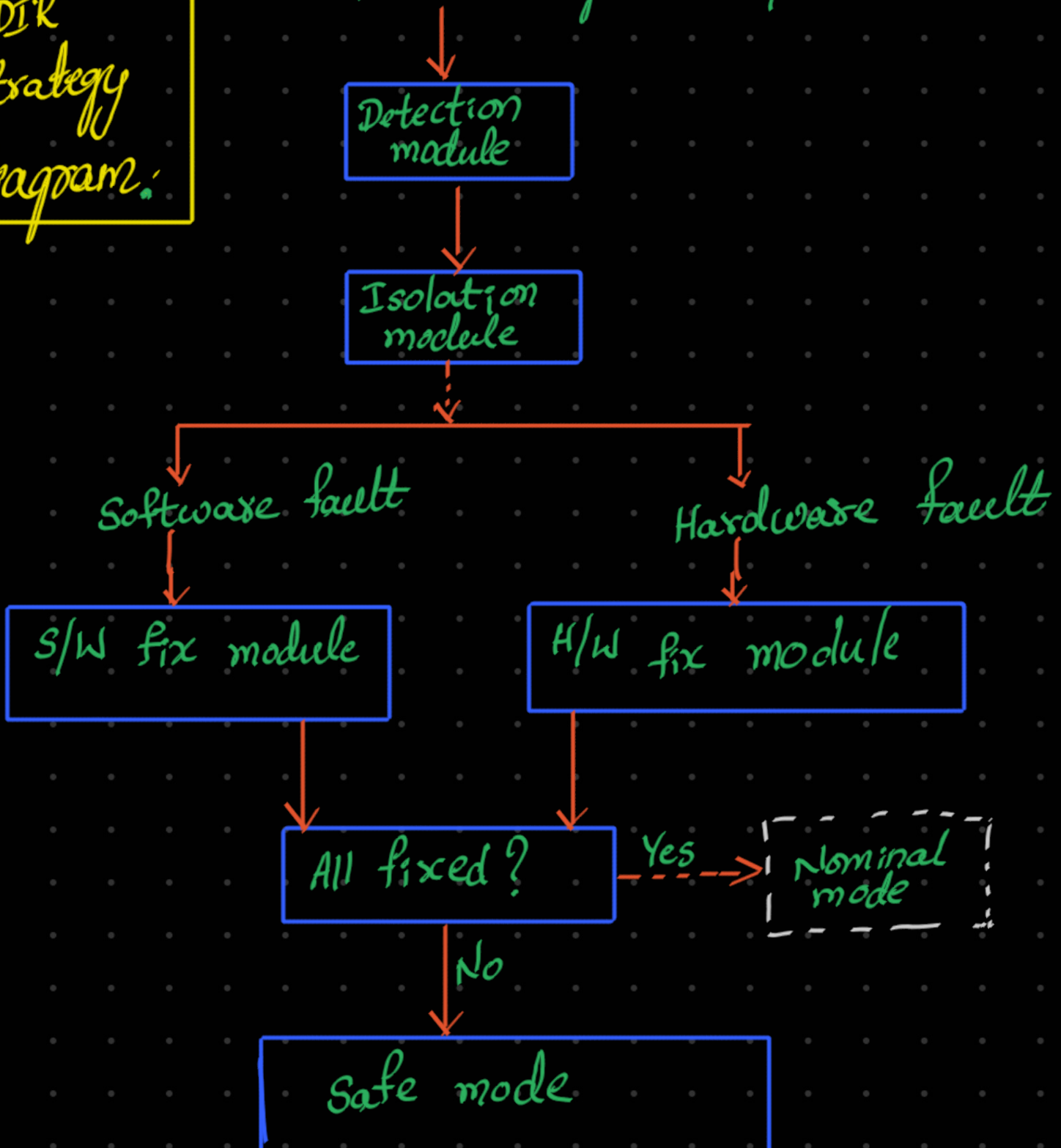
All fixed?

Yes

Nominal
mode

No

Safe mode



1] Fault detection:-

From the given problem, there are two clues to detect fault: First, if no packets arrives from the GNSS receiver for 3 seconds [receiver operates at 1Hz, meaning no signal for longer time indicate a fault], the module triggers a data timeout. Second, if CPU utilization stays $> 85\%$ for more than 5 seconds, alert triggers. The 5 second threshold is placed to prevent false alarm caused by normal CPU operation spikes.

2] Fault Isolation:-

Once fault is detected, it needs to be isolated. For this the system checks the serial interface lines to see if data is moving b/w chips. If data is actively flowing, it indicates its a software fault as receiver hardware is healthy, but computation is trapped in loop. However, if the serial lines carrying no data, then its a hardware fault because carrying no data clearly indicates the receiver is damaged.

3] Automatic recovery actions:-

Post isolating the faults, its time for automatic recovery sequence. Here, if the software fault is found, then, wipe the memory.

This process takes hardly milliseconds, thereby not blocking any running task. If hardware fault is noted, then, cut the receiver's power for few seconds, this forces it a restart power cycle thereby clearing up any induced chip lockups.

4] Entering Safe mode:-

If neither software reset nor hardware reset helps in rectifying the issue, then, the system migrates to safe mode. Generally, one software restart & two hardware restart cycle should rectify the issue. But, if GNSS receiver still remains completely inoperable, then it shutdown the operations to save limited battery. For CPU, if its clocked at 100% for some time, then, switch off of receiver will reduce the CPU load to 10%. This helps in protecting the processor from permanent damage.

5] Autonomous return to Nominal mode:-

Returning to nominal mode is allowed, iff, the issue is just a minor software fault and system reboots successfully. However, if it is a hardware fault & payload is forced

into safe mode, then, autonomous return to nominal state is not allowed. Ground operator checks the telemetry data and takes a decision.

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